

Telematics Driver Behaviour Scoring Evaluation

Expert Review Questionnaire by Matthew Bowyer

Thank you for taking the time to participate in this expert review.

The purpose of this review is to assess whether the scoring telematics output is credible, operationally useful, and aligned with real-world fleet management expectations.

All outputs shown to you are anonymised. No driver names, client names, or commercially sensitive identifiers are included.

The intention is to use expert feedback to either refine and improve the scoring framework or validate that the system meets professional fleet management standards.

What is your current Role? *

HOD: ADAS

Years in fleet/telematics industry? *

☐ 0-2

☒ 3-5

☐ 6-10

☐ 11+

What fleet type do you work with? *

- ☐ Passenger
- ☐ Light commercial
- ☐ Heavy commercial
- ☐ Mixed/Multiple fleet types

What are your approximate vehicles managed? *

- ☐ 1-50
- ☐ 51-200
- ☐ 201-1000
- ☒ 1000+

Do you currently use a driver score/rating system? *

- ☒ Yes
- ☐ No

Materials - You have been provided with three outputs to review.

A. Behaviour Category Weight Structure

The final driver behaviour score is built from six ranked measures derived from aggregated telematics data.

1. Total driving exposure (drive hours): 5%
2. Night driving ratio: 10%
3. Harsh manoeuvres (cornering, braking, acceleration): 15%
4. Speeding events: 15%
5. Power and compliance violations: 30%
6. AI-based attention and situational risk indicators: 25%

Step 1: Relative Scoring

Each vehicle is compared to all other vehicles in the population.

For every metric:

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- The score reflects where the vehicle sits relative to the group

This means the system is population-based, not threshold-based.

Step 2: Weighted Combination

The six relative scores are multiplied by their assigned weights and combined into a single UBI proxy score.

This produces one overall risk score per vehicle.

Risk Classification

Vehicles are classified into (based on distribution):

- Low Risk → Bottom third
- Medium Risk → Middle third
- High Risk → Top third

This ensures risk levels reflect relative performance within the fleet rather than arbitrary fixed cutoffs.

B. Event-to-Category Mapping Table

This table explains:

- What each event represents operationally.
- Whether the event comes from the telematics device or AI camera.
- Which behavioural category the event contributes to.

This ensures transparency in how raw events influence the final score.

Event Name	Short Description	Source Device	Affected rank column
Speeding	Vehicle speed exceeds client-set max (default 130 km/h)	Telematics	Speeding
Harsh Braking	Sudden decrease in speed	Telematics	Harsh
Harsh Acceleration	Sudden increase in speed	Telematics	Harsh
Harsh Cornering	Sharp direction change	Telematics	Harsh
Night Driving Ratio	% driving between 22:00–05:00	Telematics	Night Ratio
Seatbelt Off	Seatbelt not detected (ADC or CAN)	Telematics	Violations
Power Disconnect	External battery disconnected	Telematics	Violations
Camera Covered	Camera lens obstruction detected	AI Camera	Violations
Lane Departure	Unsafe lane deviation detected	AI Camera	Attention
Forward Collision Risk	Unsafe closing distance to vehicle ahead	AI Camera	Attention
Headway Monitoring	Unsafe following distance	AI Camera	Attention
Phone Usage	Phone use while driving	AI Camera	Attention
Driver Distraction	Visual distraction detected	AI Camera	Attention
Eye Closure	Prolonged eye closure detected	AI Camera	Attention
Fatigue	Fatigue pattern detected	AI Camera	Attention
Yawning	Yawning behaviour detected	AI Camera	Attention
Smoking	Smoking detected	AI Camera	Attention
Pedestrian Collision Risk	Pedestrian impact risk detected	AI Camera	Attention

C. Risk Ranking Tables

These include:

- Top 10 highest-risk vehicles for the month
- Top 10 lowest-risk vehicles
- A breakdown of behavioural category contributions for each vehicle

These rankings simulate how a fleet manager would:

- Prioritise coaching
- Investigate incidents
- Review driver performance

Top 10 High-Risk Vehicles

vehicle_id	drive_hours	night_ratio	harsh_total	speed_total	power_events	attention_total	ubi_proxy_score	vehicle_behaviour_class
365598883	179.511111	0.107644	139	78	81	10833	0.838276	High
338306241	202.011944	0.097927	18	28	259	18612	0.837586	High
365599055	186.381111	0.101793	29	9	121	13215	0.820259	High
454260123	146.435278	0.102366	32	24	129	11318	0.817759	High
507302529	159.878056	0.147248	18	17	176	10319	0.808793	High
307989723	133.258333	0.173131	26	2	4	9287	0.798621	High
334148753	230.435000	0.103583	116	73	2	5544	0.790517	High
333545827	483.996389	0.132929	133	2	58	198	0.784655	High
371660225	109.795833	0.018054	29	18	11	15630	0.783879	High
336854753	242.640000	0.008467	120	40	247	10563	0.779655	High

Top 10 Low-Risk Vehicles

vehicle_id	drive_hours	night_ratio	harsh_total	speed_total	power_events	attention_total	ubi_proxy_score	vehicle_behaviour_class
376677144	0.000000	0.000000	0	0	0	0	0.236379	Low
378317132	0.000000	0.000000	0	0	0	0	0.236379	Low
509636643	0.553056	0.000000	0	0	0	0	0.238017	Low
509500466	4.272500	0.000000	0	0	0	0	0.241293	Low
443664708	8.836389	0.000000	0	0	0	0	0.242155	Low
459505629	29.329444	0.000000	0	0	0	0	0.245431	Low
459503546	77.516389	0.000000	0	0	0	0	0.261983	Low
509387327	2.853611	0.000000	2	0	0	0	0.267328	Low
509637190	0.490278	0.000000	5	0	0	0	0.273793	Low
457001649	0.671111	0.031457	0	0	0	0	0.275948	Low

All questions below are based on the above information and examples. *

Do the top 10 and bottom 10 rankings look credible?

1 (Not credible) - 5 (Credible)



What (if anything) looked clearly wrong or suspicious in the ranking? *

Do not rank drivers that have no drive time.

Are any of the six behaviour categories inappropriate, overemphasised, or missing? *

1. Total driving exposure (drive hours): 5%
2. Night driving ratio: 10%
3. Harsh manoeuvres (cornering, braking, acceleration): 15%
4. Speeding events: 15%
5. Power and compliance violations: 30%
6. AI-based attention and situational risk indicators: 25%

Consider changing the %. Speeding, fatigue and distraction are some the leading causes for accidents in SA

Do the current weights for the six behaviour categories reflect real-world risk importance? *

1 (Not at all) - 5 (Perfectly)



If you could change the weights, what would you adjust and why? *

Yes, as per above comment some weights are too high (power & compliance), for me power speeding and AI should be similar

Would you trust this score to prioritise coaching, escalate investigations, or inform incentives/disciplinary decisions? *

Yes, if done correctly this will highlight severities and make it easier.

Final Remarks and Appreciation

Thank you for taking the time to complete this evaluation.

Your feedback provides essential real-world insight into whether this driver scoring framework reflects operational fleet risk and management priorities. Expert input such as yours strengthens the practical relevance and defensibility of the research.

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What is your current Role? *

SQL Reporting Manager

Years in fleet/telematics industry? *

- ☐ 0-2
- ☐ 3-5
- ☒ 6-10
- ☐ 11+

What fleet type do you work with? *

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What are your approximate vehicles managed? *

- ☐ 1-50
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Do you currently use a driver score/rating system? *

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B. Event-to-Category Mapping Table

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Do the top 10 and bottom 10 rankings look credible?

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What (if anything) looked clearly wrong or suspicious in the ranking? *

Aggregation should be applied (either time or distance). Violation counts are unreliable unless all vehicles stream data at the same interval. Speeding violations are measured based on road speeds, not taking into account vehicle-specific speed limits. Eg: A heavy commercial vehicle could be exceeding its vehicle-specific threshold, whilst still being under the posted road speed, thus not registering speed violations

Are any of the six behaviour categories inappropriate, overemphasised, or missing? *

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'Power and compliance violations' is severely overemphasized and 'Total driving exposure (drive hours)' is underemphasized. Eg: 2 drivers using the same vehicle. 1 driver drives 4 days a week, the other driver only 1 day a week. Both travel same route per day and obtain the same amount of violations per category per day. Driver 2 would be rated as a better driver due to non-aggregation and non-acknowledgement of driver utilization - even though their aggregated performances are exactly equal

Do the current weights for the six behaviour categories reflect real-world risk importance? *

1 (Not at all) - 5 (Perfectly)

1



2



3



4



5



If you could change the weights, what would you adjust and why? *

1. 15% | 2. 10% | 3. 15% | 4. 15% | 5. 10% | 6. 35%. Attention and Situational Risks are the major factors of incidents. Driver exposure was given greater weight because time behind the wheel increases risk

Would you trust this score to prioritise coaching, escalate investigations, or inform incentives/disciplinary decisions? *

Yes, as long as aggregation is applied and drivers are rated fairly against each other. Its suggested to rate performaces based on vehicle type (heavy, light commercial, etc)

Final Remarks and Appreciation

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Senior Report Write

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All questions below are based on the above information and examples. *

Do the top 10 and bottom 10 rankings look credible?

1 (Not credible) - 5 (Credible)



What (if anything) looked clearly wrong or suspicious in the ranking? *

Firstly, many of the behavioral metrics (such as harsh cornering, speeding events, power events, and attention alerts) appear to be raw event counts rather than normalized rates. Vehicles with significantly higher drive hours will naturally accumulate more events, which may incorrectly classify high-utilization vehicles as higher risk. These metrics would be more meaningful if normalized per 100 km or per driving hour.

Secondly, the low-risk ranking includes vehicles with zero or near-zero driving hours and zero events. These vehicles appear as low risk simply because they did not operate during the period, rather than demonstrating safe driving behavior. Ideally, vehicles with minimal exposure should be excluded or flagged separately to avoid skewing the ranking.

Lastly, the ranking does not appear to account for vehicle class differences. Different vehicle types (e.g., light vehicles, trucks, or specialized equipment) naturally produce different driving patterns and event frequencies. Without segmentation by vehicle class, comparisons across the fleet may not accurately reflect true driver risk.

Overall, while the structure of the scoring model is logical, these factors could distort the credibility of the rankings.

Are any of the six behaviour categories inappropriate, overemphasised, or missing? *

- 1. Total driving exposure (drive hours): 5%
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In my opinion, Total Driving Exposure (drive hours) is not appropriate to include as a scored behavioral category in a fleet management context. Exposure should rather be used to normalize behavioral metrics (e.g., events per 100 km or per driving hour) rather than contributing directly to the risk score. If a vehicle does not drive, there is effectively no behavior to evaluate.

Additionally, Power and Compliance Violations (30%) and AI-based Attention Indicators (25%) appear somewhat overemphasized, as together they account for more than half of the total score. While both are important indicators, such high weighting could disproportionately influence the overall risk score.

The remaining categories—night driving, harsh cornering, and speeding—are appropriate behavioral indicators for assessing driver risk.

Do the current weights for the six behaviour categories reflect real-world risk importance? *

1 (Not at all) - 5 (Perfectly)



If you could change the weights, what would you adjust and why? *

I would reduce the weighting of Power and Compliance Violations and increase the weighting of Night Driving. While compliance violations indicate policy breaches, they do not always directly translate into higher accident risk. Night driving, however, is inherently riskier due to reduced visibility, driver fatigue, the possibility of impaired driving, and increased security risks such as hijacking in certain areas.

Suggested weighting adjustment:

Total driving exposure: 0% (used only for normalizing behavior)

Night driving ratio: 20%

Harsh manoeuvres: 20%

Speeding events: 20%

Power and compliance violations: 20%

AI-based attention indicators: 20%

This approach places greater emphasis on actual driving conditions and behaviors that influence accident risk, while still keeping compliance and attention factors relevant.

Would you trust this score to prioritise coaching, escalate investigations, or inform incentives/disciplinary decisions? *

Yes

Final Remarks and Appreciation

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From my observation is that the scoring model would benefit from normalizing behavioral events by distance or driving hours (e.g., events per 100 km). Vehicles with higher utilization naturally generate more events, which may unfairly influence their ranking.

It may also be useful to segment scoring by vehicle class, as different vehicle types (e.g., light vehicles, trucks, or specialized equipment) operate under different conditions and produce different driving patterns.

Overall, the framework is a useful starting point for identifying behavioral trends, but incorporating normalized metrics and vehicle-type segmentation could improve the fairness and reliability of the risk scores.

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What is your current Role? *

Head: Fleet Services

Years in fleet/telematics industry? *

- ☐ 0-2
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What fleet type do you work with? *

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- Whether the event comes from the telematics device or AI camera.
- Which behavioural category the event contributes to.

This ensures transparency in how raw events influence the final score.

Event Name	Short Description	Source Device	Affected rank column
Speeding	Vehicle speed exceeds client-set max (default 130 km/h)	Telematics	Speeding
Harsh Braking	Sudden decrease in speed	Telematics	Harsh
Harsh Acceleration	Sudden increase in speed	Telematics	Harsh
Harsh Cornering	Sharp direction change	Telematics	Harsh
Night Driving Ratio	% driving between 22:00–05:00	Telematics	Night Ratio
Seatbelt Off	Seatbelt not detected (ADC or CAN)	Telematics	Violations
Power Disconnect	External battery disconnected	Telematics	Violations
Camera Covered	Camera lens obstruction detected	AI Camera	Violations
Lane Departure	Unsafe lane deviation detected	AI Camera	Attention
Forward Collision Risk	Unsafe closing distance to vehicle ahead	AI Camera	Attention
Headway Monitoring	Unsafe following distance	AI Camera	Attention
Phone Usage	Phone use while driving	AI Camera	Attention
Driver Distraction	Visual distraction detected	AI Camera	Attention
Eye Closure	Prolonged eye closure detected	AI Camera	Attention
Fatigue	Fatigue pattern detected	AI Camera	Attention
Yawning	Yawning behaviour detected	AI Camera	Attention
Smoking	Smoking detected	AI Camera	Attention
Pedestrian Collision Risk	Pedestrian impact risk detected	AI Camera	Attention

C. Risk Ranking Tables

These include:

- Top 10 highest-risk vehicles for the month
- Top 10 lowest-risk vehicles
- A breakdown of behavioural category contributions for each vehicle

These rankings simulate how a fleet manager would:

- Prioritise coaching
- Investigate incidents
- Review driver performance

Top 10 High-Risk Vehicles

vehicle_id	drive_hours	night_ratio	harsh_total	speed_total	power_events	attention_total	ubi_proxy_score	vehicle_behaviour_class
365598883	179.511111	0.107644	139	78	81	10833	0.838276	High
338306241	202.011944	0.097927	18	28	259	18612	0.837586	High
365599055	186.381111	0.101793	29	9	121	13215	0.820259	High
454260123	146.435278	0.102366	32	24	129	11318	0.817759	High
507302529	159.878056	0.147248	18	17	176	10319	0.808793	High
307989723	133.258333	0.173131	26	2	4	9287	0.798621	High
334148753	230.435000	0.103583	116	73	2	5544	0.790517	High
333545827	483.996389	0.132929	133	2	58	198	0.784655	High
371660225	109.795833	0.018054	29	18	11	15630	0.783879	High
336854753	242.640000	0.008467	120	40	247	10563	0.779655	High

Top 10 Low-Risk Vehicles

vehicle_id	drive_hours	night_ratio	harsh_total	speed_total	power_events	attention_total	ubi_proxy_score	vehicle_behaviour_class
376677144	0.000000	0.000000	0	0	0	0	0.236379	Low
378317132	0.000000	0.000000	0	0	0	0	0.236379	Low
509636643	0.553056	0.000000	0	0	0	0	0.238017	Low
509500466	4.272500	0.000000	0	0	0	0	0.241293	Low
443664708	8.836389	0.000000	0	0	0	0	0.242155	Low
459505629	29.329444	0.000000	0	0	0	0	0.245431	Low
459503546	77.516389	0.000000	0	0	0	0	0.261983	Low
509387327	2.853611	0.000000	2	0	0	0	0.267328	Low
509637190	0.490278	0.000000	5	0	0	0	0.273793	Low
457001649	0.671111	0.031457	0	0	0	0	0.275948	Low

All questions below are based on the above information and examples. *

Do the top 10 and bottom 10 rankings look credible?

1 (Not credible) - 5 (Credible)

1



2



3



4



5



What (if anything) looked clearly wrong or suspicious in the ranking? *

No

Are any of the six behaviour categories inappropriate, overemphasised, or missing? *

1. Total driving exposure (drive hours): 5%
2. Night driving ratio: 10%
3. Harsh manoeuvres (cornering, braking, acceleration): 15%
4. Speeding events: 15%
5. Power and compliance violations: 30%
6. AI-based attention and situational risk indicators: 25%

Power and compliance violations - % ranking to high

Do the current weights for the six behaviour categories reflect real-world risk importance? *

1 (Not at all) - 5 (Perfectly)

1



2



3



4



5



If you could change the weights, what would you adjust and why? *

I would bring down the 'Power and compliance violations' % as there are cases where it could not be driver related issue. I would be more severe on the Harsh manoeuvres as that is likely to cause wear and tear, accidents and fuel usage.

Would you trust this score to prioritise coaching, escalate investigations, or inform incentives/disciplinary decisions? *

I would trust the data to take both actions to put a incentive in place to reward good driver. and to coach and up skill the drivers at risk before it is to late.

Final Remarks and Appreciation

Thank you for taking the time to complete this evaluation.

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Speeding violations should take into account how long the vehicle has been speeding, not just the occurrence of the event. Scoring should only apply once the vehicle has been above the speed limit for a defined period of time. This allows for a short window of over-speeding that may occur during legitimate manoeuvres, such as overtaking.

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Google Forms

Telematics Driver Behaviour Scoring Evaluation

Expert Review Questionnaire by Matthew Bowyer

Thank you for taking the time to participate in this expert review.

The purpose of this review is to assess whether the scoring telematics output is credible, operationally useful, and aligned with real-world fleet management expectations.

All outputs shown to you are anonymised. No driver names, client names, or commercially sensitive identifiers are included.

The intention is to use expert feedback to either refine and improve the scoring framework or validate that the system meets professional fleet management standards.

What is your current Role? *

Operations Manager

Years in fleet/telematics industry? *

☐ 0-2

☒ 3-5

☐ 6-10

☐ 11+

What fleet type do you work with? *

- ☐ Passenger
- ☐ Light commercial
- ☐ Heavy commercial
- ☒ Mixed/Multiple fleet types

What are your approximate vehicles managed? *

- ☐ 1-50
- ☐ 51-200
- ☒ 201-1000
- ☐ 1000+

Do you currently use a driver score/rating system? *

- ☒ Yes
- ☐ No

Materials - You have been provided with three outputs to review.

A. Behaviour Category Weight Structure

The final driver behaviour score is built from six ranked measures derived from aggregated telematics data.

1. Total driving exposure (drive hours): 5%
2. Night driving ratio: 10%
3. Harsh manoeuvres (cornering, braking, acceleration): 15%
4. Speeding events: 15%
5. Power and compliance violations: 30%
6. AI-based attention and situational risk indicators: 25%

Step 1: Relative Scoring

Each vehicle is compared to all other vehicles in the population.

For every metric:

- Performance is converted into a value between 0 and 1
- The score reflects where the vehicle sits relative to the group

This means the system is population-based, not threshold-based.

Step 2: Weighted Combination

The six relative scores are multiplied by their assigned weights and combined into a single UBI proxy score.

This produces one overall risk score per vehicle.

Risk Classification

Vehicles are classified into (based on distribution):

- Low Risk → Bottom third
- Medium Risk → Middle third
- High Risk → Top third

This ensures risk levels reflect relative performance within the fleet rather than arbitrary fixed cutoffs.

B. Event-to-Category Mapping Table

This table explains:

- What each event represents operationally.
- Whether the event comes from the telematics device or AI camera.
- Which behavioural category the event contributes to.

This ensures transparency in how raw events influence the final score.

Event Name	Short Description	Source Device	Affected rank column
Speeding	Vehicle speed exceeds client-set max (default 130 km/h)	Telematics	Speeding
Harsh Braking	Sudden decrease in speed	Telematics	Harsh
Harsh Acceleration	Sudden increase in speed	Telematics	Harsh
Harsh Cornering	Sharp direction change	Telematics	Harsh
Night Driving Ratio	% driving between 22:00–05:00	Telematics	Night Ratio
Seatbelt Off	Seatbelt not detected (ADC or CAN)	Telematics	Violations
Power Disconnect	External battery disconnected	Telematics	Violations
Camera Covered	Camera lens obstruction detected	AI Camera	Violations
Lane Departure	Unsafe lane deviation detected	AI Camera	Attention
Forward Collision Risk	Unsafe closing distance to vehicle ahead	AI Camera	Attention
Headway Monitoring	Unsafe following distance	AI Camera	Attention
Phone Usage	Phone use while driving	AI Camera	Attention
Driver Distraction	Visual distraction detected	AI Camera	Attention
Eye Closure	Prolonged eye closure detected	AI Camera	Attention
Fatigue	Fatigue pattern detected	AI Camera	Attention
Yawning	Yawning behaviour detected	AI Camera	Attention
Smoking	Smoking detected	AI Camera	Attention
Pedestrian Collision Risk	Pedestrian impact risk detected	AI Camera	Attention

C. Risk Ranking Tables

These include:

- Top 10 highest-risk vehicles for the month
- Top 10 lowest-risk vehicles
- A breakdown of behavioural category contributions for each vehicle

These rankings simulate how a fleet manager would:

- Prioritise coaching
- Investigate incidents
- Review driver performance

Top 10 High-Risk Vehicles

vehicle_id	drive_hours	night_ratio	harsh_total	speed_total	power_events	attention_total	ubi_proxy_score	vehicle_behaviour_class
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333545827	483.996389	0.132929	133	2	58	198	0.784655	High
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Top 10 Low-Risk Vehicles

vehicle_id	drive_hours	night_ratio	harsh_total	speed_total	power_events	attention_total	ubi_proxy_score	vehicle_behaviour_class
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509636643	0.553056	0.000000	0	0	0	0	0.238017	Low
509500466	4.272500	0.000000	0	0	0	0	0.241293	Low
443664708	8.836389	0.000000	0	0	0	0	0.242155	Low
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509637190	0.490278	0.000000	5	0	0	0	0.273793	Low
457001649	0.671111	0.031457	0	0	0	0	0.275948	Low

All questions below are based on the above information and examples. *

Do the top 10 and bottom 10 rankings look credible?

1 (Not credible) - 5 (Credible)



What (if anything) looked clearly wrong or suspicious in the ranking? *

Less driving hours does not constitute low risk. To class a vehicle as low with not enough driving hours could be interpreted wrong. The high risk vehicles with enough data points would appear to be more accurate.

Are any of the six behaviour categories inappropriate, overemphasised, or missing? *

1. Total driving exposure (drive hours): 5%
2. Night driving ratio: 10%
3. Harsh manoeuvres (cornering, braking, acceleration): 15%
4. Speeding events: 15%
5. Power and compliance violations: 30%
6. AI-based attention and situational risk indicators: 25%

1. Driving exposure: Appropriate
 2. Night driving ratio: Slightly overemphasised.
 3. Harsh manoeuvres: Appropriate.
 4. Speeding events: Appropriate
 5. Power and compliance violations is overemphasised.
 6. AI based attention: Inappropriate
-

Do the current weights for the six behaviour categories reflect real-world risk importance? *

1 (Not at all) - 5 (Perfectly)

1



2



3



4



5



If you could change the weights, what would you adjust and why? *

Night driving ratio: Reduce weight. Some businesses have nightshifts.

Power and compliance violations: Reduce weight. Typical large commercial organisations have vehicles regularly have breakdowns or will go on maintenance where power events occurs. Some vehicles are also fitted with isolation switches causing power events.

Camera obstruction events could also be caused due to the placement of cameras. Sun visors and sunlight often causes this event to trigger.

AI Based attention: This category should be sub-categorised/split. Some events should be weighted more than others. Fatigue, Cell phone use and eyes closed are often considered more serious offences than a person yawning or smoking.

Would you trust this score to prioritise coaching, escalate investigations, or inform incentives/disciplinary decisions? *

Yes

Final Remarks and Appreciation

Thank you for taking the time to complete this evaluation.

Your feedback provides essential real-world insight into whether this driver scoring framework reflects operational fleet risk and management priorities. Expert input such as yours strengthens the practical relevance and defensibility of the research.

If you have any additional comments or reflections not covered in the questions above, please include them here:

To be taken into account: AI Cameras will trigger true AND false events which will have an impact on the scores.

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Telematics Driver Behaviour Scoring Evaluation

Expert Review Questionnaire by Matthew Bowyer

Thank you for taking the time to participate in this expert review.

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All outputs shown to you are anonymised. No driver names, client names, or commercially sensitive identifiers are included.

The intention is to use expert feedback to either refine and improve the scoring framework or validate that the system meets professional fleet management standards.

What is your current Role? *

Video Solutions - global

Years in fleet/telematics industry? *



0-2



3-5



6-10



11+

What fleet type do you work with? *

- ☐ Passenger
- ☐ Light commercial
- ☐ Heavy commercial
- ☒ Mixed/Multiple fleet types

What are your approximate vehicles managed? *

- ☐ 1-50
- ☐ 51-200
- ☐ 201-1000
- ☒ 1000+

Do you currently use a driver score/rating system? *

- ☒ Yes
- ☐ No

Materials - You have been provided with three outputs to review.

A. Behaviour Category Weight Structure

The final driver behaviour score is built from six ranked measures derived from aggregated telematics data.

1. Total driving exposure (drive hours): 5%
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Risk Classification

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This ensures risk levels reflect relative performance within the fleet rather than arbitrary fixed cutoffs.

B. Event-to-Category Mapping Table

This table explains:

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Pedestrian Collision Risk	Pedestrian impact risk detected	AI Camera	Attention

C. Risk Ranking Tables

These include:

- Top 10 highest-risk vehicles for the month
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These rankings simulate how a fleet manager would:

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- Investigate incidents
- Review driver performance

Top 10 High-Risk Vehicles

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Top 10 Low-Risk Vehicles

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459505629	29.329444	0.000000	0	0	0	0	0.245431	Low
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509637190	0.490278	0.000000	5	0	0	0	0.273793	Low
457001649	0.671111	0.031457	0	0	0	0	0.275948	Low

All questions below are based on the above information and examples. *

Do the top 10 and bottom 10 rankings look credible?

1 (Not credible) - 5 (Credible)



What (if anything) looked clearly wrong or suspicious in the ranking? *

307989723 seems to have too view issues for its ranking. 457001649 In my experience, if there are zero harsh / negative events, the configs are simply set wrong. The vehicle being on for approximately 30 minutes could also mean that the vehicle was idling. Including distance travelled could be relevant.

Are any of the six behaviour categories inappropriate, overemphasised, or missing? *

1. Total driving exposure (drive hours): 5%
2. Night driving ratio: 10%
3. Harsh manoeuvres (cornering, braking, acceleration): 15%
4. Speeding events: 15%
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Real power violations should exclude the device completely. It is a very real issue, but multiple power events typically mean that the vehicle itself / the device itself is faulty.

Do the current weights for the six behaviour categories reflect real-world risk importance? *

1 (Not at all) - 5 (Perfectly)



If you could change the weights, what would you adjust and why? *

Reduce the ratio for power. High power events mean that the device / vehicle has a bad battery

Would you trust this score to prioritise coaching, escalate investigations, or inform incentives/disciplinary decisions? *

Within the correct configurations and weights, yes

Final Remarks and Appreciation

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What is your current Role? *

SATF Telematic Manager

Years in fleet/telematics industry? *



0-2



3-5



6-10



11+

What fleet type do you work with? *

- ☒ Passenger
- ☐ Light commercial
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What are your approximate vehicles managed? *

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- ☐ 201-1000
- ☒ 1000+

Do you currently use a driver score/rating system? *

- ☒ Yes
- ☐ No

Materials - You have been provided with three outputs to review.

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443664708	8.836389	0.000000	0	0	0	0	0.242155	Low
459505629	29.329444	0.000000	0	0	0	0	0.245431	Low
459503546	77.516389	0.000000	0	0	0	0	0.261983	Low
509387327	2.853611	0.000000	2	0	0	0	0.267328	Low
509637190	0.490278	0.000000	5	0	0	0	0.273793	Low
457001649	0.671111	0.031457	0	0	0	0	0.275948	Low

All questions below are based on the above information and examples. *

Do the top 10 and bottom 10 rankings look credible?

1 (Not credible) - 5 (Credible)



What (if anything) looked clearly wrong or suspicious in the ranking? *

One suspicious aspect is that some vehicles ranked as high risk appear to have relatively low speeding or harsh driving events, suggesting their ranking may be influenced disproportionately by factors like very high drive hours or unusually large AI attention counts, while some low-risk vehicles have zero drive hours and events but still receive non-zero risk scores, which can seem counterintuitive.

Are any of the six behaviour categories inappropriate, overemphasised, or missing? *

1. Total driving exposure (drive hours): 5%
2. Night driving ratio: 10%
3. Harsh manoeuvres (cornering, braking, acceleration): 15%
4. Speeding events: 15%
5. Power and compliance violations: 30%
6. AI-based attention and situational risk indicators: 25%

Overall the categories are appropriate, but power and compliance violations (30%) and AI-based attention indicators (25%) may be slightly overemphasised, as together they account for 55% of the total score, which could dominate the ranking, while other important behaviours such as following distance or fatigue indicators might be missing from the model.

Do the current weights for the six behaviour categories reflect real-world risk importance? *

1 (Not at all) - 5 (Perfectly)

1



2



3



4



5



If you could change the weights, what would you adjust and why? *

I would slightly reduce power and compliance violations from 30% to about 25% and AI-based attention indicators from 25% to about 20%, and redistribute that weight to speeding and harsh manoeuvres, since speeding and aggressive driving are strongly linked to real-world crash risk and should have slightly more influence on the score.

Would you trust this score to prioritise coaching, escalate investigations, or inform incentives/disciplinary decisions? *

Yes, but primarily as a screening or prioritisation tool rather than the sole decision-maker, meaning it would be useful to identify high-risk drivers for coaching or review, while still verifying the underlying events before making disciplinary or incentive decisions.

Final Remarks and Appreciation

Thank you for taking the time to complete this evaluation.

Your feedback provides essential real-world insight into whether this driver scoring framework reflects operational fleet risk and management priorities. Expert input such as yours strengthens the practical relevance and defensibility of the research.

If you have any additional comments or reflections not covered in the questions above, please include them here:

Overall, the scoring framework appears logical and transparent, especially with the use of relative scoring and clear event-to-category mapping, which helps explain how the final risk score is produced. One potential improvement would be ensuring that extreme outliers in certain metrics (such as very high drive hours or AI attention counts) do not disproportionately influence the final ranking. Additionally, it may be useful to periodically review the weights and validate them against real incident or crash data to confirm that the model continues to reflect real-world risk accurately.

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Telematics Driver Behaviour Scoring Evaluation

Expert Review Questionnaire by Matthew Bowyer

Thank you for taking the time to participate in this expert review.

The purpose of this review is to assess whether the scoring telematics output is credible, operationally useful, and aligned with real-world fleet management expectations.

All outputs shown to you are anonymised. No driver names, client names, or commercially sensitive identifiers are included.

The intention is to use expert feedback to either refine and improve the scoring framework or validate that the system meets professional fleet management standards.

What is your current Role? *

CVAT Manager

Years in fleet/telematics industry? *



0-2



3-5



6-10



11+

What fleet type do you work with? *

- ☐ Passenger
- ☐ Light commercial
- ☐ Heavy commercial
- ☒ Mixed/Multiple fleet types

What are your approximate vehicles managed? *

- ☒ 1-50
- ☐ 51-200
- ☐ 201-1000
- ☐ 1000+

Do you currently use a driver score/rating system? *

- ☐ Yes
- ☒ No

Materials - You have been provided with three outputs to review.

A. Behaviour Category Weight Structure

The final driver behaviour score is built from six ranked measures derived from aggregated telematics data.

1. Total driving exposure (drive hours): 5%
2. Night driving ratio: 10%
3. Harsh manoeuvres (cornering, braking, acceleration): 15%
4. Speeding events: 15%
5. Power and compliance violations: 30%
6. AI-based attention and situational risk indicators: 25%

Step 1: Relative Scoring

Each vehicle is compared to all other vehicles in the population.

For every metric:

- Performance is converted into a value between 0 and 1
- The score reflects where the vehicle sits relative to the group

This means the system is population-based, not threshold-based.

Step 2: Weighted Combination

The six relative scores are multiplied by their assigned weights and combined into a single UBI proxy score.

This produces one overall risk score per vehicle.

Risk Classification

Vehicles are classified into (based on distribution):

- Low Risk → Bottom third
- Medium Risk → Middle third
- High Risk → Top third

This ensures risk levels reflect relative performance within the fleet rather than arbitrary fixed cutoffs.

B. Event-to-Category Mapping Table

This table explains:

- What each event represents operationally.
- Whether the event comes from the telematics device or AI camera.
- Which behavioural category the event contributes to.

This ensures transparency in how raw events influence the final score.

Event Name	Short Description	Source Device	Affected rank column
Speeding	Vehicle speed exceeds client-set max (default 130 km/h)	Telematics	Speeding
Harsh Braking	Sudden decrease in speed	Telematics	Harsh
Harsh Acceleration	Sudden increase in speed	Telematics	Harsh
Harsh Cornering	Sharp direction change	Telematics	Harsh
Night Driving Ratio	% driving between 22:00–05:00	Telematics	Night Ratio
Seatbelt Off	Seatbelt not detected (ADC or CAN)	Telematics	Violations
Power Disconnect	External battery disconnected	Telematics	Violations
Camera Covered	Camera lens obstruction detected	AI Camera	Violations
Lane Departure	Unsafe lane deviation detected	AI Camera	Attention
Forward Collision Risk	Unsafe closing distance to vehicle ahead	AI Camera	Attention
Headway Monitoring	Unsafe following distance	AI Camera	Attention
Phone Usage	Phone use while driving	AI Camera	Attention
Driver Distraction	Visual distraction detected	AI Camera	Attention
Eye Closure	Prolonged eye closure detected	AI Camera	Attention
Fatigue	Fatigue pattern detected	AI Camera	Attention
Yawning	Yawning behaviour detected	AI Camera	Attention
Smoking	Smoking detected	AI Camera	Attention
Pedestrian Collision Risk	Pedestrian impact risk detected	AI Camera	Attention

C. Risk Ranking Tables

These include:

- Top 10 highest-risk vehicles for the month
- Top 10 lowest-risk vehicles
- A breakdown of behavioural category contributions for each vehicle

These rankings simulate how a fleet manager would:

- Prioritise coaching
- Investigate incidents
- Review driver performance

Top 10 High-Risk Vehicles

vehicle_id	drive_hours	night_ratio	harsh_total	speed_total	power_events	attention_total	ubi_proxy_score	vehicle_behaviour_class
365598883	179.511111	0.107644	139	78	81	10833	0.838276	High
338306241	202.011944	0.097927	18	28	259	18612	0.837586	High
365599055	186.381111	0.101793	29	9	121	13215	0.820259	High
454260123	146.435278	0.102366	32	24	129	11318	0.817759	High
507302529	159.878056	0.147248	18	17	176	10319	0.808793	High
307989723	133.258333	0.173131	26	2	4	9287	0.798621	High
334148753	230.435000	0.103583	116	73	2	5544	0.790517	High
333545827	483.996389	0.132929	133	2	58	198	0.784655	High
371660225	109.795833	0.018054	29	18	11	15630	0.783879	High
336854753	242.640000	0.008467	120	40	247	10563	0.779655	High

Top 10 Low-Risk Vehicles

vehicle_id	drive_hours	night_ratio	harsh_total	speed_total	power_events	attention_total	ubi_proxy_score	vehicle_behaviour_class
376677144	0.000000	0.000000	0	0	0	0	0.236379	Low
378317132	0.000000	0.000000	0	0	0	0	0.236379	Low
509636643	0.553056	0.000000	0	0	0	0	0.238017	Low
509500466	4.272500	0.000000	0	0	0	0	0.241293	Low
443664708	8.836389	0.000000	0	0	0	0	0.242155	Low
459505629	29.329444	0.000000	0	0	0	0	0.245431	Low
459503546	77.516389	0.000000	0	0	0	0	0.261983	Low
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457001649	0.671111	0.031457	0	0	0	0	0.275948	Low

All questions below are based on the above information and examples. *

Do the top 10 and bottom 10 rankings look credible?

1 (Not credible) - 5 (Credible)



What (if anything) looked clearly wrong or suspicious in the ranking? *

N/A

Are any of the six behaviour categories inappropriate, overemphasised, or missing? *

1. Total driving exposure (drive hours): 5%
2. Night driving ratio: 10%
3. Harsh manoeuvres (cornering, braking, acceleration): 15%
4. Speeding events: 15%
5. Power and compliance violations: 30%
6. AI-based attention and situational risk indicators: 25%

No

Do the current weights for the six behaviour categories reflect real-world risk importance? *

1 (Not at all) - 5 (Perfectly)



If you could change the weights, what would you adjust and why? *

1. I think it should be excluded.
 5. Should be reduced a little.
 6. Should be increased.
-

Would you trust this score to prioritise coaching, escalate investigations, or inform incentives/disciplinary decisions? *

I would trust it for all 3

Final Remarks and Appreciation

Thank you for taking the time to complete this evaluation.

Your feedback provides essential real-world insight into whether this driver scoring framework reflects operational fleet risk and management priorities. Expert input such as yours strengthens the practical relevance and defensibility of the research.

If you have any additional comments or reflections not covered in the questions above, please include them here:

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